

High School Mathematics: An Intuitive Introduction

By LinuxChad628

To the GOAT Mr. P (You know who you are)

Introduction

The goal of this book is to deliver an intuitive introduction to high-school level mathematics. My goal here is not to replace high school, but rather to provide you with the necessary mathematical background to do well in your classes. As for how you should read this book, that depends on your comfort level when it comes to mathematics. If you are new to this type of math, you should read slowly, and make sure to do all of the exercises. On the other hand, if you are comfortable with math, you can go faster and only do the exercises that you think you need extra practice with. However, no matter your level of expertise, I would recommend reading the entire thing and not skipping any of the book. This is because the goal of this book is not only to teach you math, but also to introduce a new way of looking at this subject, one that focuses on intuition first, rather than raw computation.

Contents:

Unit 1: Algebra I

- Chapter 1: Expressions
- Chapter 2: Graphs of Lines
- Chapter 3: Solving Linear Equations
- Chapter 4: Systems of Equations
- Chapter 5: Graphing 1-D and 2-D Inequalities
- Chapter 6: Solving Inequalities
- Chapter 7: The Quadratic
- Chapter 8: Solving the Quadratic By Graphing
- Chapter 9: Solving the Quadratic By Factorization
- Chapter 10: Perfect Square Quadratics
- Chapter 11: Completing the Square
- Chapter 12: The General Quadratic and the Quadratic Formula
- Chapter 13: Absolute Value
- Chapter 14: Absolute Value Equations and Inequalities

Unit 2: Geometry

- Chapter 15: Preliminary Notions of Geometry
- Chapter 16: Proof
- Chapter 17: Line Segments and Angles
- Chapter 18: Parallel Lines
- Chapter 19: Triangles
- Chapter 20: Polygons
- Chapter 21: Transformations
- Chapter 22: Extending the Real Numbers
- Chapter 23: Transformations in the Complex Plane
- Chapter 24: Trigonometry
- Chapter 25: Circles

Unit 3: Precalculus

- Chapter 26: Exploring Functions